

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2023/2024

DECEMBER EXAMINATION

UCCD2103 OPERATING SYSTEMS

FRIDAY, 29 DECEMBER 2023

TIME: 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING

Instructions to Candidates:

This question paper consists of **THREE (3)** questions in Section A and **TWO (2)** questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.

Should a candidate answer more than **ONE (1)** question in **Section B**, marks will only be awarded for the **FIRST** question in that section in the order the candidate submits the answers.

Each question carries 25 marks.

Candidates are allowed to use calculator.

Answer questions only in the answer booklet provided.

UCCD2103 OPERATING SYSTEMS**Section A: Answer ALL questions.**

- Q1. (a) Referring to the following sequential scenarios, define each state transition of Process A based on the 7-state model:
- (i) Process A is created and dispatched to execute immediately. (1 mark)
 - (ii) Process A then waits for user input via a touch-screen display. (1 mark)
 - (iii) While waiting for that event to occur, another process (Process B) is dispatched to execute and immediately consume a large amount of memory. Process A is then suspended to virtual memory. (1 mark)
 - (iv) The user touched the display as an input to Process A. (1 mark)
 - (v) Process A is then selected to resume execution. (1 mark)
 - (vi) Process A then quits. (1 mark)
- (b) Figure 1.1 shows the CPU and I/O utilisations of a uniprocessor computer system when Process C and Process D are executing. Process C has CPU and I/O instructions whilst Process D has only CPU instructions with a total of 15ms CPU service time.

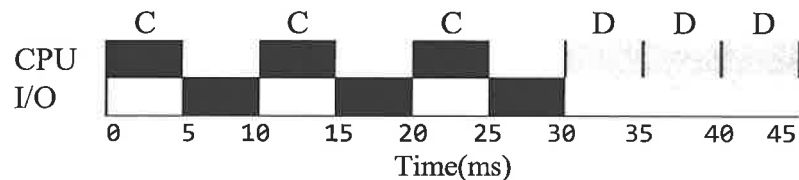


Figure 1.1: CPU and I/O utilisations of a uniprocessor computer system

Assume Process C and Process D are in READY state at $t = 0$ ms and Process C is selected to execute first. Answer the following questions:

- (i) Name the system that matches the execution pattern shown in Figure 1.1. Briefly explain the main disadvantage of the system. (3 marks)
- (ii) What are the turnaround times for Process C and Process D shown in Figure 1.1? (4 marks)
- (iii) The main disadvantage explained in Q1(b)(i) can be overcome using multiprogramming system. Draw the execution pattern of Process C and Process D execute on a uniprocessor multiprogramming system and then calculate their turnaround times. (6 marks)

UCCD2103 OPERATING SYSTEMS**Q1. (Continued)**

- (c) Identify the type of real-time task (hard or soft) in the application or system mentioned in the following statements and briefly justify your answer:
- (i) Advanced Driver-Assistance Systems (ADAS) are automotive technology for vehicle collision avoidance and driver aids. Sensors such as LiDAR (Light Detection and Ranging) and camera are used for emergency braking, pedestrian detection, and lane departure warning. (3 marks)
 - (ii) Real-time Operating System (OS) deployed on mixed reality Augmented Reality (AR) / Virtual Reality (VR) headsets. (3 marks)
- [Total: 25 marks]

- Q2. (a) This question is about the Completely Fair Scheduler (CFS) in the Linux kernel. Assume that the system has a uniprocessor and the period for CFS is set at 20 milliseconds. Figure 2.1 shows the distribution of CPU time among three CPU-bound processes - Process E, Process F and Process G. They are in Ready state at $t = 0$ milliseconds. Other scheduling parameters at $t = 0$ milliseconds are shown in Table 2.1:

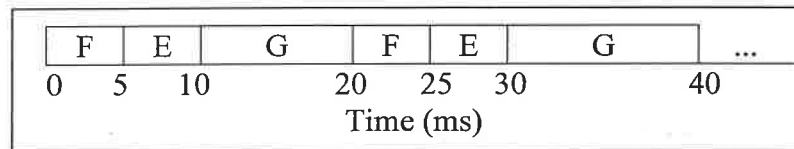


Figure 2.1: CPU utilizations of Process E, Process F and Process G

Table 2.1: Parameters for Process E, Process F and Process G

Parameters	Process E	Process F	Process G
Virtual runtime (nanoseconds)	115,412,123	110,412,123	120,412,123
Nice value	X	Y	Z

The minimum virtual runtime of the red-black tree is 110,412,123 nanoseconds. Using the information provided in Appendix A (refer to page 9), answer the following questions:

- (i) Determine the values of X, Y, and Z in Table 2.1. (3 marks)
- (ii) Calculate the virtual runtimes of Process E, Process F and Process G at $t = 40$ milliseconds. (9 marks)
- (iii) How can the Process F receive the most CPU allocation among these three processes? (2 marks)

UCCD2103 OPERATING SYSTEMS**Q2. (Continued)**

- (b) Assume that a computer's main memory has a total of 6 frames. Frame #0, frame #3, and frame #4 are assigned to Process B and frame #1, frame #2, and frame #5 are assigned to Process A. Figure 2.2 shows the current state of the main memory where frame #4 and frame #5 are empty.

frame #0	B 8
frame #1	A 0
frame #2	A 4
frame #3	B 3
frame #4	
frame #5	

Figure 2.2: Current state of memory frames

The sequence of page requests arrival is A0, B8, A4, B3. Then, the following page trace occurs (process name and its respective page number):

A1, A2, B2, B8, B7, A0, A5, B4, B7, B4

Fixed Allocation and Local Replacement Scope are used. Show the steps of each page replacement performed as following:

- (i) Least Recently Used (LRU) policy is used. (5 marks)
- (ii) Clock replacement policy is used. Indicate the use-bit in your answer. Assume that the use-bit in frame #0 and frame #1 in Figure 2.2 are set to 1 and frame #2, frame #3, frame #4, and frame #5 are set to 0. The clock arrows for Process A and Process B are now pointing at frame #2 and frame #3 respectively. (6 marks)

[Total: 25 marks]

- Q3. (a) Consider a disk head is now positioned at track 180 and has completed a previous request at track 220. The requests of track are in the following sequence:

245, 230, 125, 90, 175, 150, 200, 190

Draw the seek pattern on a graph using the C-SCAN disk scheduling algorithm starting from track 180. Let the x-axis and y-axis represents the request number and track number. (8 marks)

- (b) Disk cache is the buffer in main memory that stores a copy of some sectors from disk. Explain the impact to the performance of an application that requires a lot of disk I/O operations if the disk cache size is reduced by half. (5 marks)

UCCD2103 OPERATING SYSTEMS**Q3. (Continued)**

- (c) Consider a disk with 20 blocks (Block 0 to Block 19) and its bit tables is given as follow:

10010101100011001100

Given that 8 blocks are required to store File B and the searching for free blocks will start from the first block, Block 0. Assume that *Indexed Allocation with Variable Length Portions* was implemented and the first available block will be used as the Index Block

- (i) Draw the file allocation table with the entry for File B. (2 marks)
- (ii) List the contents of the Index Block. (5 marks)
- (d) Consider a very large file to be stored in a disk with sufficient free blocks. If *Index Allocation* was used, explain the performance in terms of the operation duration in the following operations:
- (i) Storing the file. (2 marks)
- (ii) Reading the file. (3 marks)

[Total: 25 marks]

UCCD2103 OPERATING SYSTEMS**Section B: Answer ANY ONE (1) question.**

- Q4. (a) Table 4.1 shows the current allocation of resources to each process, the priority associated with each process and the resources that are still needed by each process to proceed and complete their execution. Smaller priority value indicates higher priority.
- The resources currently available are $R1 = 1$, $R2 = 0$, $R3 = 0$ and $R4 = 1$. Based on the information in Table 4.1 and the resources available, all the processes are deadlocked. Resolve the deadlock by *successively abort deadlocked processes with the lowest priority*. Show the resources available after each abortion in your workings. (9 marks)

Table 4.1: Resources allocation of processes with priority value

Process	Priority value	Allocated				Needed			
		R1	R2	R3	R4	R1	R2	R3	R4
A	4	1	0	0	2	2	2	2	2
B	2	2	1	3	2	4	1	2	3
C	5	1	1	0	0	2	2	3	4
D	3	1	1	2	1	2	3	1	3
E	1	1	1	2	3	6	3	3	6

- (b) Given a user process of 8GB, page frame size is 4KB and each page table entry is 2 bytes, calculate the size of root page table in two-level scheme. Show all your calculations. (4 marks)
- (c) Table 4.2 shows a 16-bit virtual address format whereas Table 4.3 shows a portion of a segment table. Translate the following virtual addresses to their corresponding physical address. Show all the relevant values in your answer.

Table 4.2: 16-bit virtual address format

4 bits	12 bits
Segment number	Offset

Table 4.3: Portion of a segment table

Segment number	Base	Length (decimal)
11	0000 0100 0010 0110	1024
12	0000 0100 0010 0001	1536
13	0000 0011 0000 0111	2048

- (i) 1101011111100100 (3 marks)
- (ii) 1011010000001000 (3 marks)
- (d) (i) In the paging approach, a virtual memory reference can cause two page faults. Briefly explain how the two page faults occur. (4 marks)

UCCD2103 OPERATING SYSTEMS**Q4. (d) (Continued)**

- (ii) Briefly explain how the Translation Lookaside Buffer (TLB) reduce the page fault cause by a virtual memory reference. (2 marks)

[Total: 25 marks]

- Q5. (a) Given that a system has 1MB of memory available, apply the Buddy system allocation to the followings:

NOTE: Answer can be shown in the form of binary tree.

- (i) Memory requests in the sequence shown in Table 5.1. (5 marks)

Table 5.1: Memory requests

Process	Request
A	90 KB
B	100 KB
C	110 KB
D	200 KB
E	30 KB
F	116 KB
G	10 KB

- (ii) The allocations to the process E, process G, process D and process F are released. (2 marks)

- (b) Figure 5.1 shows the pseudocode of two processes with two semaphores. Illustrate a scenario demonstrating how the two processes could become deadlocked. State the changes of the semaphores' value in the scenario. You may use the line number (L1, L2, etc.) in your answer. (7 marks)

semaphore n=0, a=1;	
P1	P2
L1 {	{
L2 semWait(a);	semWait(a);
L3 //critical section	semWait(n):
L4 semSignal(n);	//critical section
L5 semSignal(a);	semSignal(a);
L6 }	}

Figure 5.1: Pseudocode of process P1 and process P2

UCCD2103 OPERATING SYSTEMS**Q5. (Continued)**

- (c) Figure 5.2 shows the skeletons of two processes, namely *Loader* and *Launcher*. *Launcher* will launch the item after the *Loader* has placed an item on the launch platform. The *Loader* will place an item on the launch platform when the platform is empty. Both processes will run iteratively.

Loader(){	Launcher(){
...	...
//load item	//launch item
...	...
}	}

Figure 5.2: Skeletons of *Loader* and *Launcher* processes

- (i) What is the minimum number of semaphore required to synchronise the execution of the two processes? (1 mark)
- (ii) Write the pseudocode for both processes in Figure 5.2. Initialise each semaphore with the appropriate value. (7 marks)
- (d) What are the **TWO (2)** differences between semaphore and mutex? (3 marks)
- [Total: 25 marks]

UCCD2103 OPERATING SYSTEMS**APPENDIX A:****Linux Completely Fair Scheduler (CFS):**

1. Tasks' time slice allocation:

$$f_i(t) = \frac{w_i}{\sum w_{rbtree}} \times p$$

$f_i(t)$ = time slice received by task i at time t ,
 w_i = weight of task i

$\sum w_{rbtree}$ = Total weight of tasks in red-black tree

p = period

2. Virtual runtime of task i in n -th iteration:

$$v_i(n) = v_i(n-1) + \left(\frac{\Delta e}{w_i} \right) \times w_{nice0}$$

$v_i(n-1)$ = virtual runtime of task i in previous iteration

Δe = elapsed time task i spent utilizing the processor during iteration n

w_i = weight of task i

w_{nice0} = weight assigned to nice 0 task = 1024

3. Mapping of nice value to weight:

```
static const int prio_to_weight[40] = {
    /* -20 */ 88761,    71755,    56483,    46273,    36291,
    /* -15 */ 29154,    23254,    18705,    14949,    11916,
    /* -10 */ 9548,     7620,     6100,     4904,     3906,
    /* -5  */ 3121,     2501,     1991,     1586,     1277,
    /* 0   */ 1024,     820,      655,      526,      423,
    /* 5   */ 335,      272,      215,      172,      137,
    /* 10  */ 110,      87,       70,       56,       45,
    /* 15  */ 36,       29,       23,       18,       15,
}
```

